

[CSA04]

Assessment Tool-1

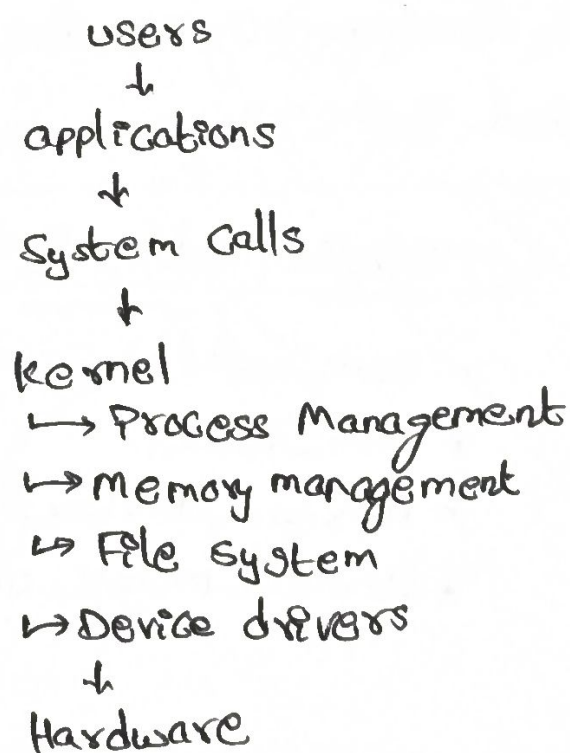
[Operating Systems]

- ① Suppose a modern operating system supports multiple users and processes with shared resources like CPU, memory, and I/O devices.

a) Types of OS

- * Multi-user OS → supports multiple users simultaneously.
- * Multi-tasking OS → runs multiple processes at the same time.
- * Time-sharing OS → shares CPU fairly among users.

b) OS structure



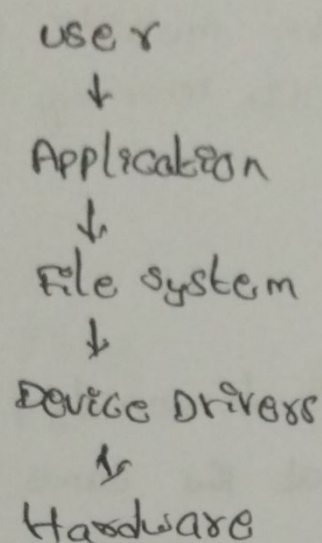
c) System calls

$\text{fork}() \rightarrow \text{exec}() \rightarrow \text{wait}() \rightarrow \text{exit}()$

② a) File & Device System calls

- * File :- $\text{open}()$, $\text{read}()$, $\text{write}()$, $\text{close}()$
- * Device :- $\text{ioctl}()$, $\text{read}()$, $\text{write}()$, $\text{close}()$

b) Layered structure



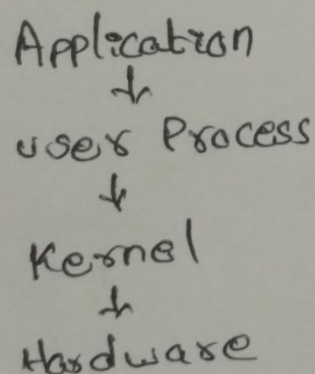
c)) C/ System calls

```
fd = open("file.txt", O_RDONLY);
read(fd, buffer, 100);
close(fd);
```

3) Type of OS

a) Real-time Operating System (RTOS) - gives fast and predictable response for embedded systems.

b) structure

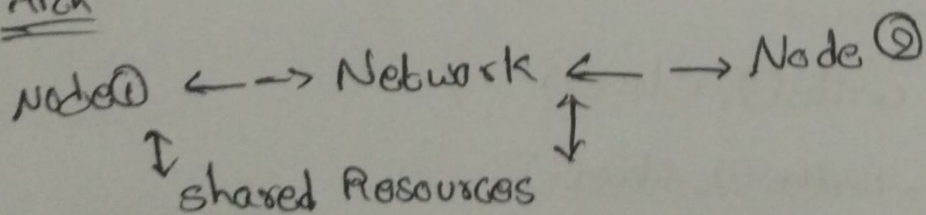


(c) Synchronization System Calls

- * sem_wait()
- * sem_post()
- * pthread_mutex_lock()
- * pthread_mutex_unlock()

④ a) Distributed Operating System

Diff:- Distributed OS uses multiple networked computers,
Centralized OS uses one central computer.

b) Arch

© IPC System Calls

```

socket() → bind() → listen()
                                     ↓
recv() ← send() ← accept()
      ↓
      close()

```